Project 1

Jooyoung Yoo

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Introduction

A Gradient Boosting model predicts a used car's value retention ratio with a test R^2 of 0.82 and an average error of about 7% of MSRP (about $2,075), and mileage and age explain most of the depreciation.

The goal of this project was to build a full machine learning pipeline that predicts how much of its original price a used car keeps, and to find which factors drive depreciation the most. The target is the value retention ratio:

Value retention ratio = current listing price / original MSRP

A ratio of 0.60 means the car is listed for 60% of what it cost new. I scraped listings from two sites (Autolist and Craigslist), collected trim-level MSRP from J.D. Power, merged everything into one dataset of 1,541 cars, engineered features, trained three regression models, and analyzed which features mattered most.

1. Data Collection

I collected 1,913 raw listings from two independent sources (954 from Autolist and 959 from Craigslist), plus 2,514 trim-level MSRP prices from J.D. Power.

Source	Type	How the data was collected	Raw rows
Autolist	National listing site, mostly dealer cars	Its JSON search API, 4 pages of 20 cars per model	954
Craigslist	Classifieds, private owners and dealers	Search pages in 10 large cities, then each listing page for specs and the description	959 (478 owner, 481 dealer)
J.D. Power	MSRP reference	One page per make, model and model year, 2008 to 2026	2,514 trim prices

Makes and models. I chose 4 makes with 3 models each, all popular cars so both sources would have enough listings:

- Toyota: Camry, Corolla, RAV4
- Honda: Civic, Accord, CR-V
- Ford: F-150, Escape, Explorer
- Chevrolet: Silverado 1500, Equinox, Malibu

Attributes. The integrated dataset keeps 18 original attributes from the sources: price, model year, make, model, trim, mileage, body type, transmission, drivetrain, fuel type, cylinders, exterior color, condition, title status, seller type, state, listing date and the listing description.

Scraping approach. Most major listing sites blocked automated requests, so I tested 14 sites and used the ones that allowed access (see Challenges in Section 7). The scrapers waited 3 seconds between requests, checked robots.txt, saved results after every search so they could resume without re-downloading, and stopped automatically on a 403 or 429 error.

2. Data Preprocessing and Integration

After cleaning and removing duplicates, 1,541 listings remained: 821 from Autolist and 720 from Craigslist, both above the 500-per-source minimum.

Cleaning step	Autolist	Craigslist	Total
Raw combined	954	959	1,913
Removed wrong models (e.g. Corolla Cross, Silverado 2500)	954	948	1,902
Removed bad prices (missing, $0 placeholders, under $1,000)	851	899	1,750
Kept model years 2008 to 2026 (years with MSRP data)	835	736	1,571
Removed bad mileage (missing, over 400k, fake "under 100")	822	735	1,557
Removed "parts only" titles	822	734	1,556
Removed duplicates (5 same VIN, 4 same description)	821	726	1,547
Removed extreme ratios (under 0.05 or over 1.5)	821	720	1,541

Unifying the two sources. The sources describe the same cars in different ways, so I had to resolve these differences before merging:

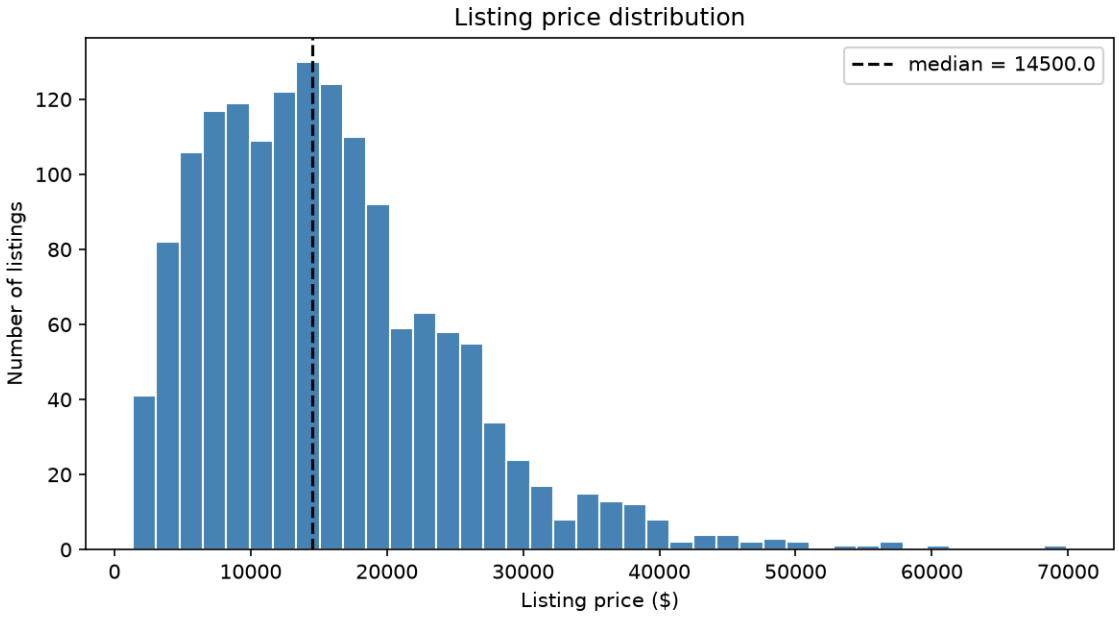
- **Column names:** `odometer` vs `mileage`, `type` vs `body_type`, `drive` vs `drivetrain`.
- **Formats:** Craigslist prices are text ("$6,900") while Autolist prices are numbers.
- **Category labels:** "grey" vs "gray", "gasoline" and "flex fuel" vs "gas", "4 cylinders" vs 4.0. Craigslist has no separate AWD option, so AWD, 4WD and 4x4 were combined into one `awd/4wd` category.
- **Location:** Craigslist gives a city and Autolist gives a state, so cities were converted to states.
- **Model names:** Craigslist model names are free text ("Chevy", "crv", or just "Toyota"), so they were matched to the 12 target models.

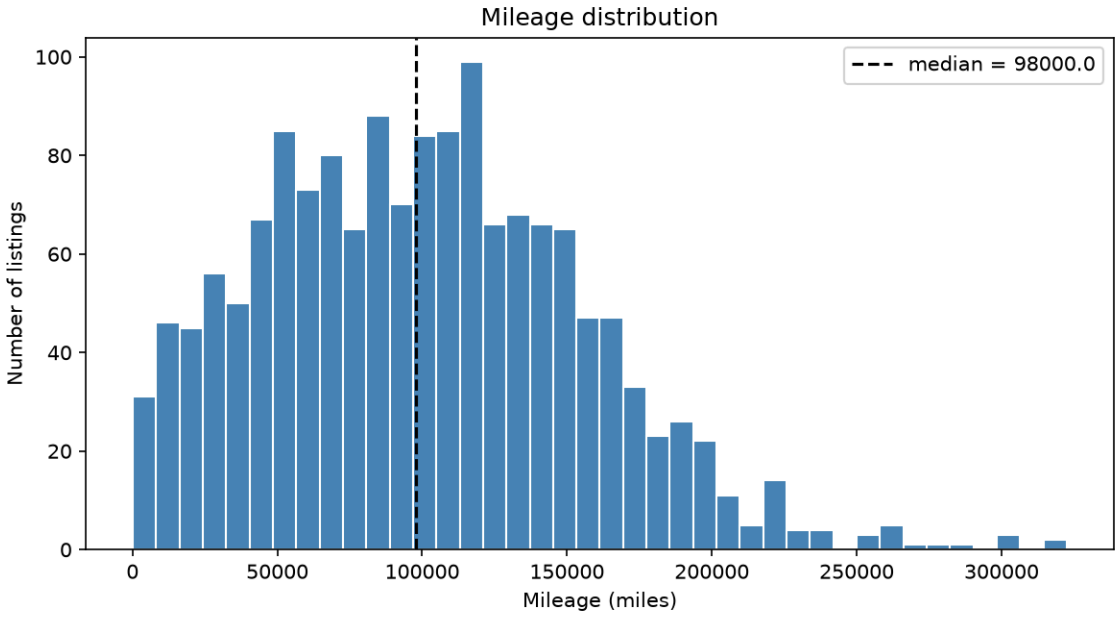
Merging MSRP. Listing trim names rarely match J.D. Power's names exactly ("LE" vs "Sedan 4D LE I4"), so trims were matched on whole words. 1,087 cars matched a specific trim. The other 454 used the median trim MSRP for that make, model and year. The value retention ratio was then computed as price divided by MSRP.

3. Data Exploration

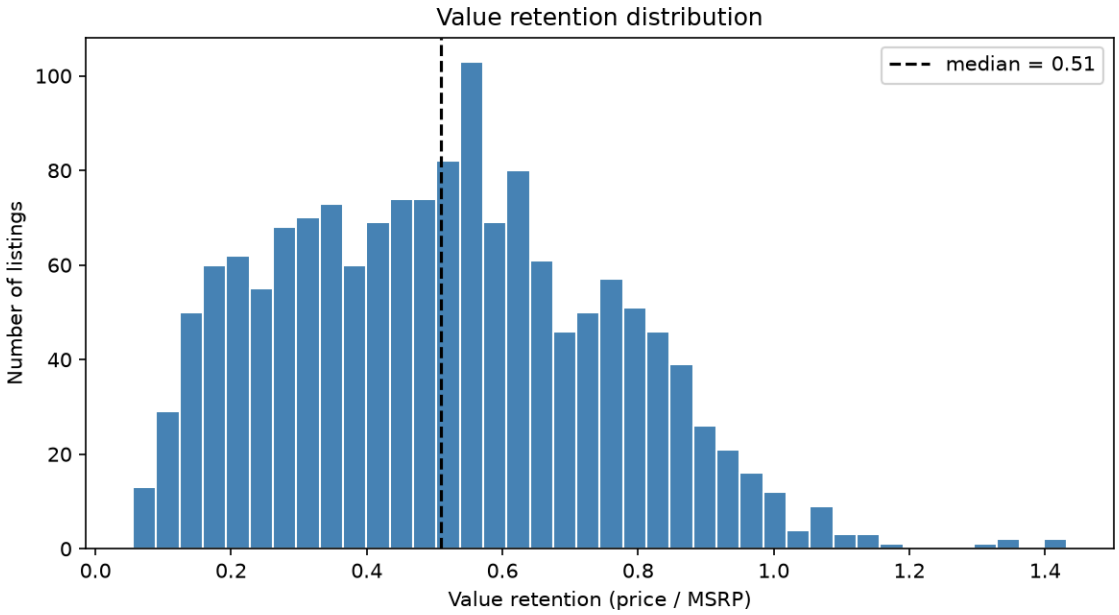
The typical car in the dataset sells for about half its original MSRP (median ratio 0.51), with a median price of $14,500, median mileage of 98,000 and median model year 2018.

Price and mileage. Prices range from $1,350 to $69,900 and are right-skewed, with most cars between $9,000 and $20,000. Mileage is spread widely, from nearly new cars to over 300,000 miles.

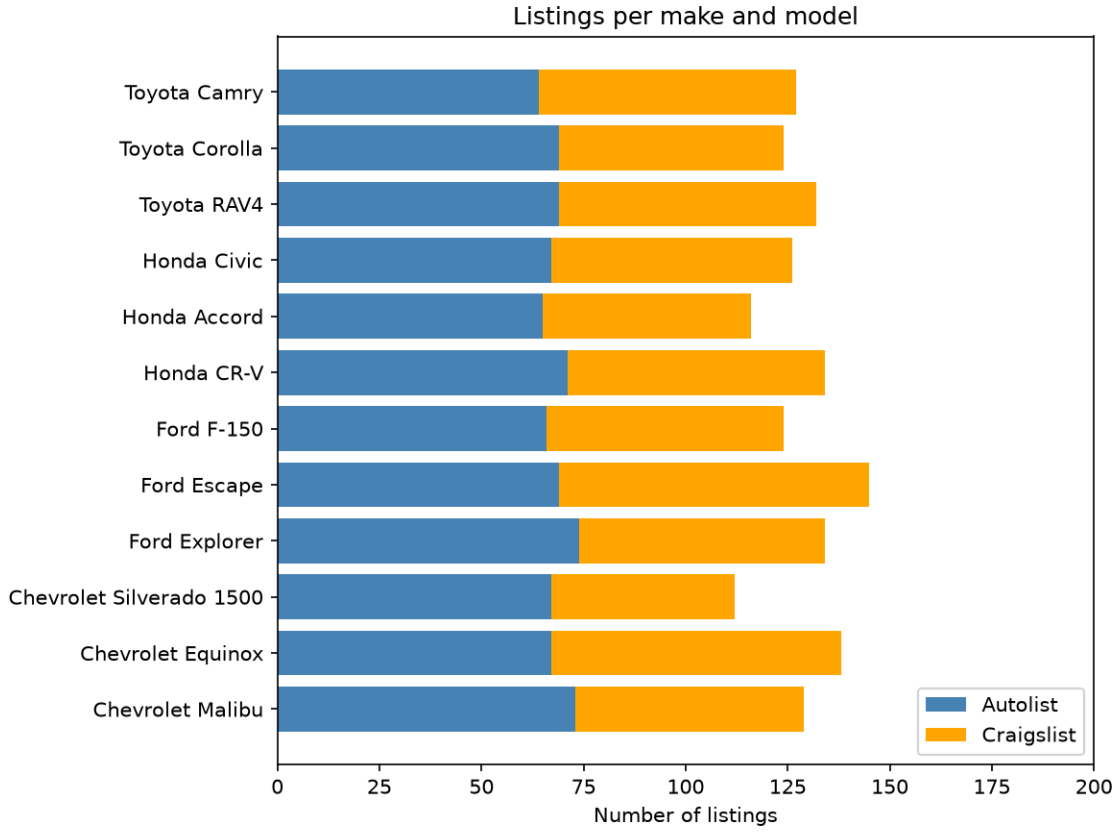


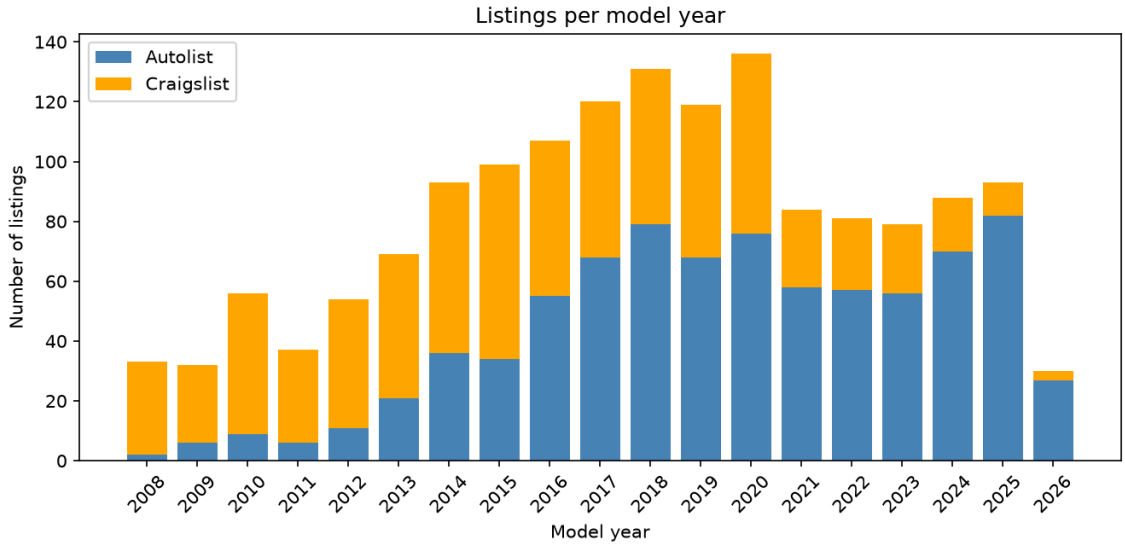


Value retention. The target ranges from 0.06 to 1.43. Ratios above 1 come from special versions (like the GR Corolla or RAV4 Prime) compared against the regular model's MSRP.



Make and model balance. Each of the 12 models has between 112 and 145 listings, and each is split roughly in half between the two sources.

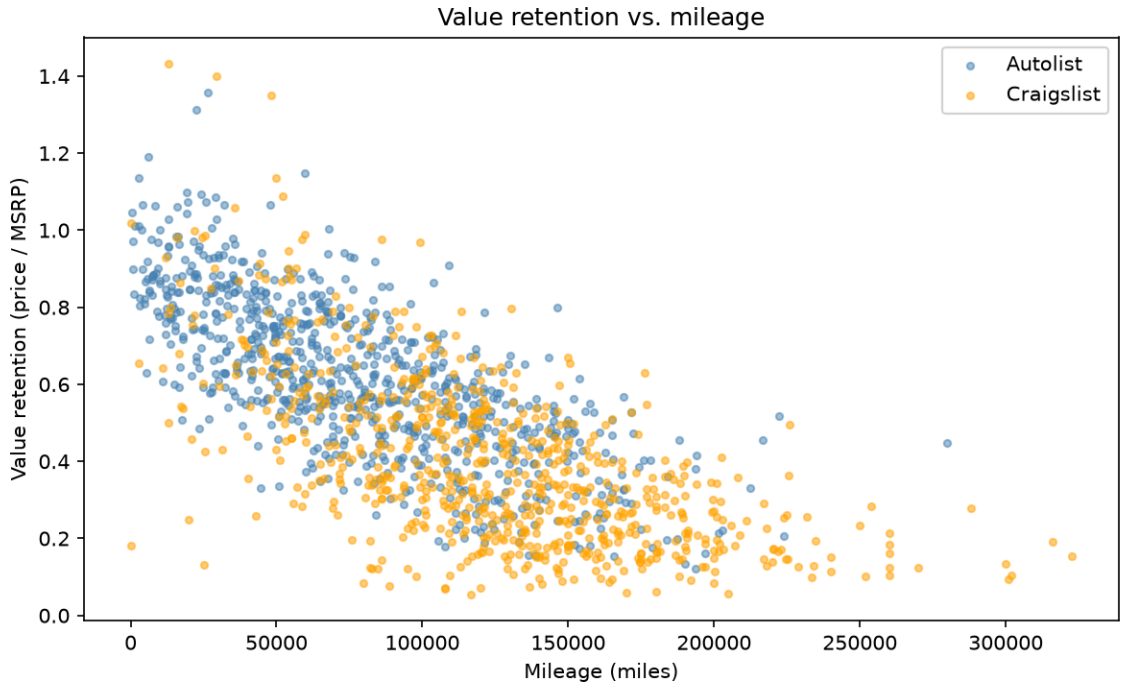


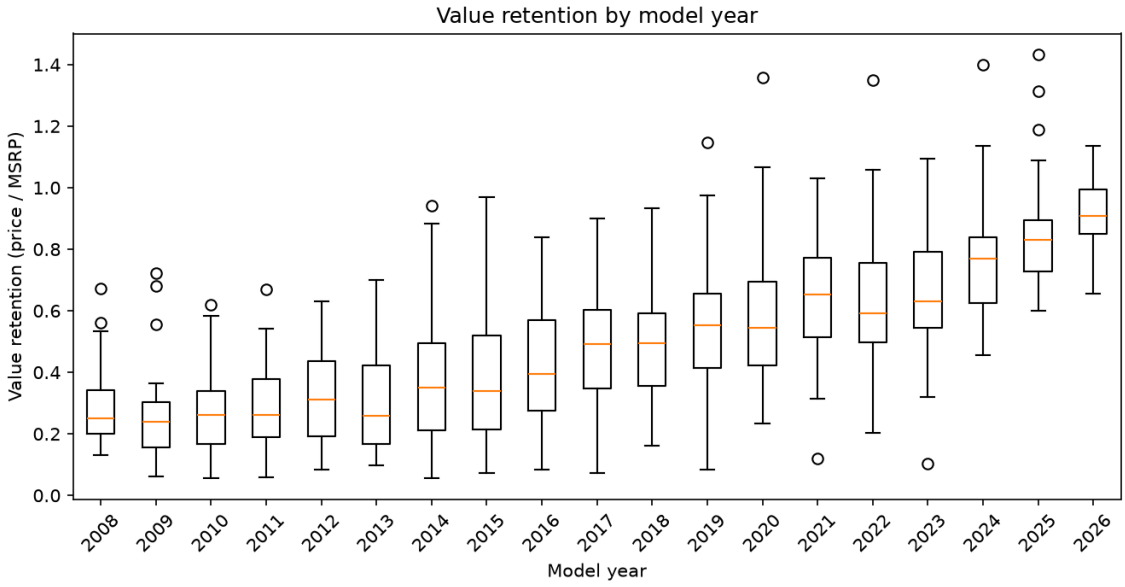


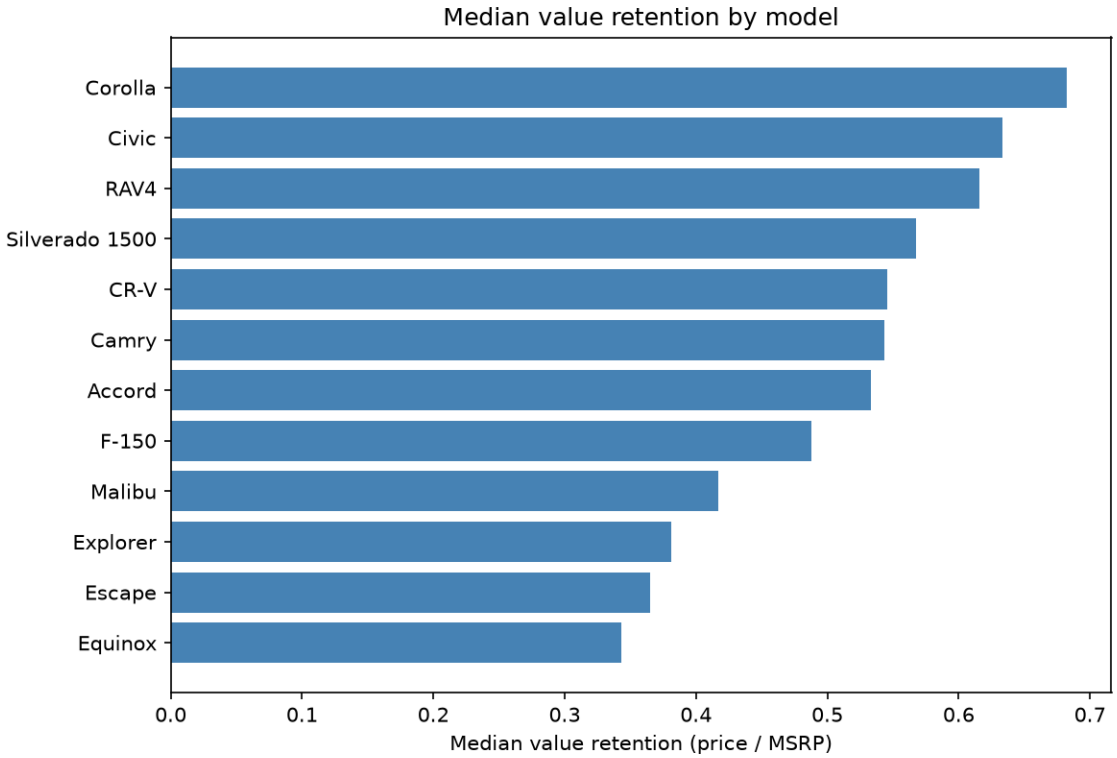
The two sources contain different kinds of cars. Craigslist cars are about four years older and have about 48,000 more miles than Autolist cars, so they retain much less of their value.

	Autolist	Craigslist
Median model year	2020	2016
Median mileage	73,000	121,000
Median price	$17,800	$9,600
Median value retention	0.61	0.35

Relationships with the target. Mileage has the strongest correlation with value retention (-0.73), followed by model year (+0.68). MSRP (+0.15) and cylinders (-0.07) are weak.


Value retention vs. mileage
Autolist
Craigslist
Value retention (price / MSRP)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
50000
100000
150000
200000
250000
300000
Mileage (miles)


Value retention by model year
Value retention (price / MSRP)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Model year



Missing data. Condition (69% missing), title status (54%) and description (53%) only exist for Craigslist listings. Exterior color (15%), drivetrain (11%) and cylinders (11%) also have gaps.

4. Feature Engineering

The final feature set has 56 numeric columns with no missing values, split 80/20 into 1,233 training cars and 308 test cars (random seed 42).

Engineered numeric features.

- `vehicle_age`: the year the car was listed minus its model year.
- `miles_per_year`: mileage divided by age (at least 1 year), which shows whether a car was driven more or less than usual for its age.
- `branded_title`: 1 if the title is rebuilt, salvage or has a lien.

Features from listing descriptions. Only Craigslist listings have descriptions, so these were extracted with keyword matching. The table shows how often each appears in Craigslist listings.

Feature	Example phrases	Craigslist listings
`desc_clean_title`	"clean title"	34.3%
`desc_maintained`	"service records", "well maintained", "new tires"	16.0%
`desc_accident_free`	"no accidents", "clean Carfax"	13.1%
`desc_one_owner`	"one owner", "1 owner"	11.1%

Feature	Example phrases	Craigslist listings
desc_needs_work	"needs work", "check engine", "as is"	7.5%
desc_premium_count	leather, sunroof, backup camera (0 to 3)	average 0.61

Missing values. Missing cylinders were filled with the most common value for that model. Fuel type and transmission (very few missing) were filled with the most common value. Drivetrain, condition and color, which have many missing values, were given their own "unknown" category. Rare body types and colors were grouped into "other".

One-hot encoding was used for low-cardinality categories (make, model, body type, drivetrain, fuel type, transmission, color, condition, seller type and source), dropping the first category of each to avoid redundant columns.

High-cardinality feature: state. State has 49 values. One-hot encoding it would add 48 sparse columns, and many states have fewer than 10 cars, which risks overfitting. I used two encodings instead:

- **Frequency encoding:** each state is replaced by the share of training cars from that state. It doesn't use the target, so it can't leak information.
- **Target encoding with smoothing:** each state is replaced by the average value retention of training cars from that state, pulled toward the overall average when a state has few cars. With n cars in the state and a smoothing weight of 10:

Encoded state = (n × state mean + 10 × overall mean) / (n + 10)

Both encodings were computed from training rows only, so the test set doesn't leak into the features. On the test set, the target encoding correlates much more with value retention (0.33) than the frequency encoding (-0.13). Part of this comes from the sources: the lowest-retention states (IL, NY, GA, PA, TX) are the Craigslist cities, which have older cars.

5. Model Development

Gradient Boosting had the best cross-validation score (R^2 0.837) and predicts value retention on the test set with an average error of 0.069, about $2,075 per car.

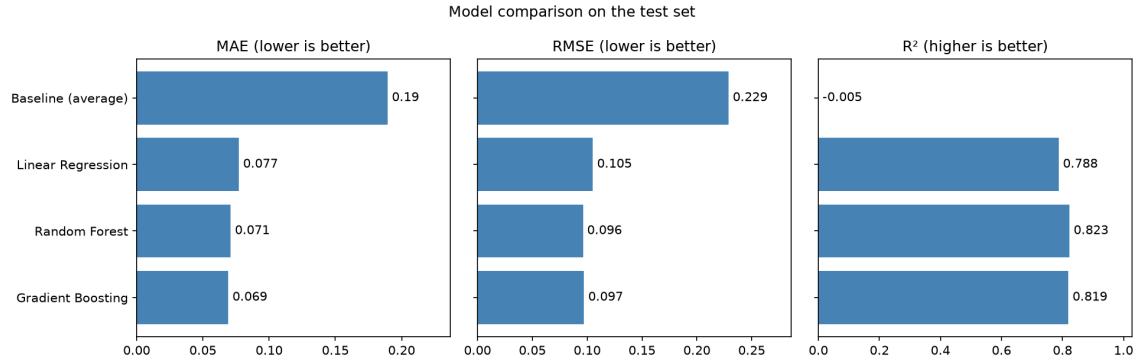
Models. I trained three regression algorithms and compared them to a baseline that always predicts the average:

- **Linear Regression** (with standardized features): one coefficient per feature, easy to interpret.
- **Random Forest:** averages 300 decision trees, each leaf holding at least 2 cars.
- **Gradient Boosting:** adds 300 small trees (depth 3) one at a time, each correcting the previous trees' errors, with a learning rate of 0.05.

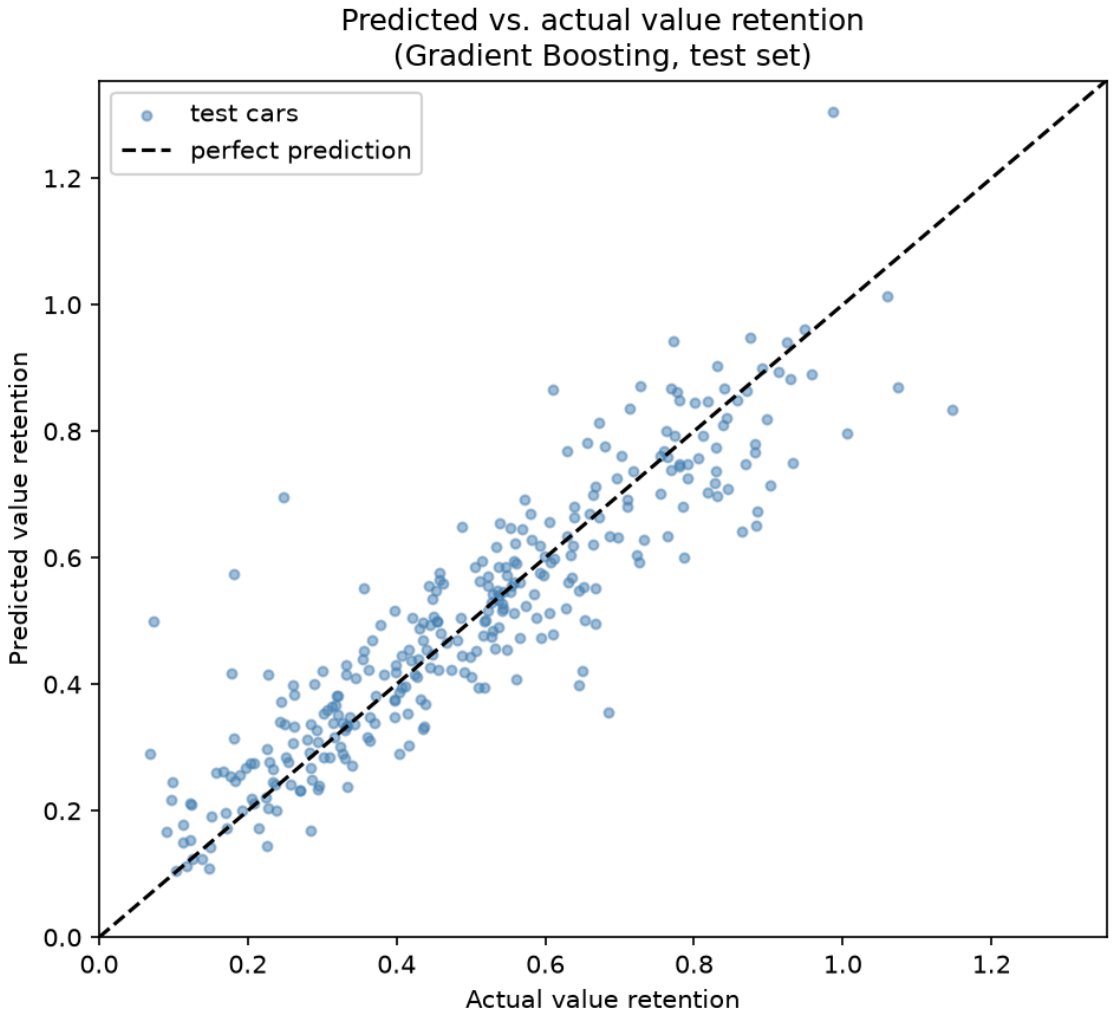
Evaluation strategy. Each model was scored with 5-fold cross-validation on the training set, then trained on all training rows and scored once on the held-out test set. The best model was chosen by cross-validation score, so the test set was only used to report the final result. Models were evaluated with MAE, RMSE and R^2. Since the target is a ratio, an MAE of 0.07 means

predictions are off by 7% of MSRP on average. I also converted the error to dollars by multiplying it by each car's MSRP.

Model	CV R^2 (± std)	Test MAE	Test RMSE	Test R^2	Test MAE ($)
Baseline (average)	-0.001	0.190	0.229	-0.005	$5,621
Linear Regression	0.821 (0.026)	0.077	0.105	0.788	$2,309
Random Forest	0.803 (0.031)	0.071	0.096	0.823	$2,124
Gradient Boosting	**0.837 (0.028)**	**0.069**	0.097	0.819	**$2,075**



Comparison. All three models cut the baseline error by more than half. Random Forest and Gradient Boosting are effectively tied on the test set: their R^2 differs by 0.004, which is smaller than the cross-validation standard deviation. Linear Regression is close behind, which suggests the relationship is mostly linear in age and mileage. The tree models gain a little by capturing curves, such as depreciation slowing down for older cars.

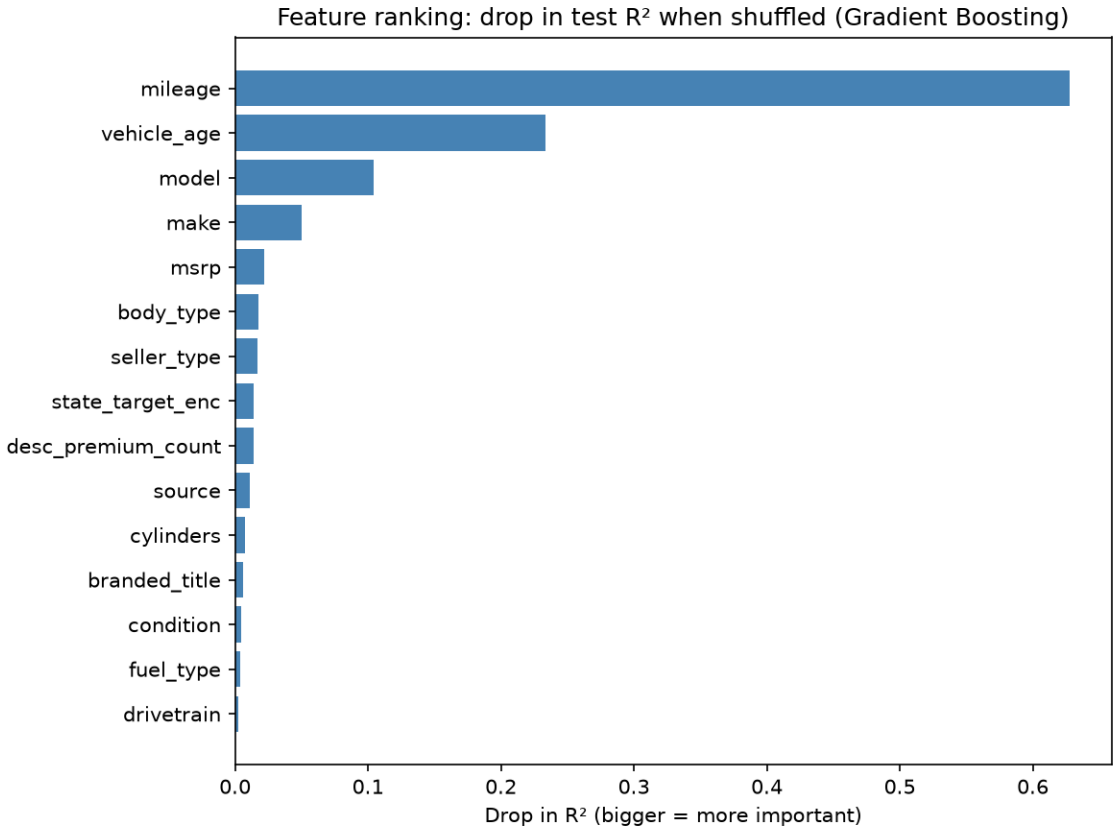


Most test predictions fall close to the diagonal. The largest errors are special versions with ratios above 1 and very cheap Craigslist listings.

6. Depreciation Analysis and Feature Ranking

Mileage is by far the most important feature, followed by vehicle age, model and make. Together, mileage and age account for most of the model's accuracy.

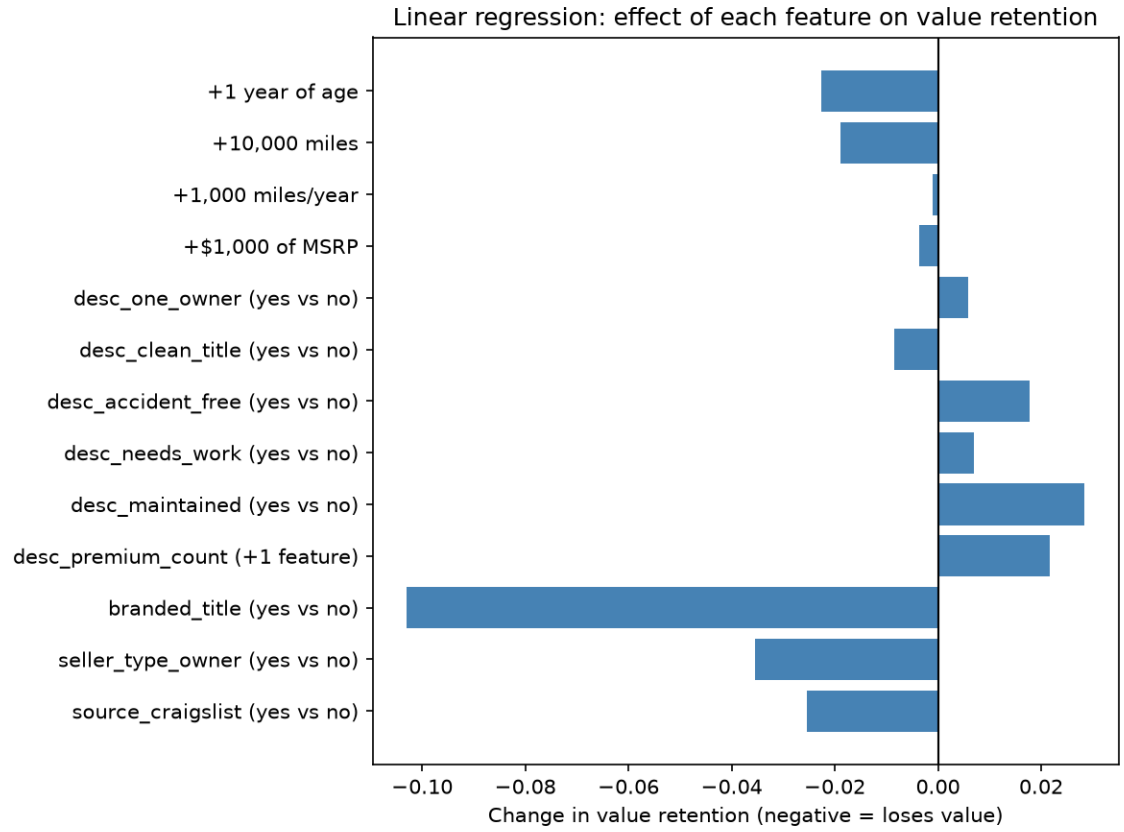
Grouped permutation importance. To rank features, I shuffled each feature on the test set and measured how much the Gradient Boosting model's R^2 dropped. A bigger drop means the model relies on that feature more. One-hot columns that belong together (for example, all 11 model columns) were shuffled together as one group, so a category isn't split across many small columns. Each shuffle was repeated 10 times.



Shuffling mileage drops R² by 0.63 and shuffling age drops it by 0.23. Model (0.10) and make (0.05) come next. MSRP, body type, seller type, the state encoding and the premium-feature count each matter a little (0.01 to 0.02). The individual description phrases and exterior color contribute almost nothing.

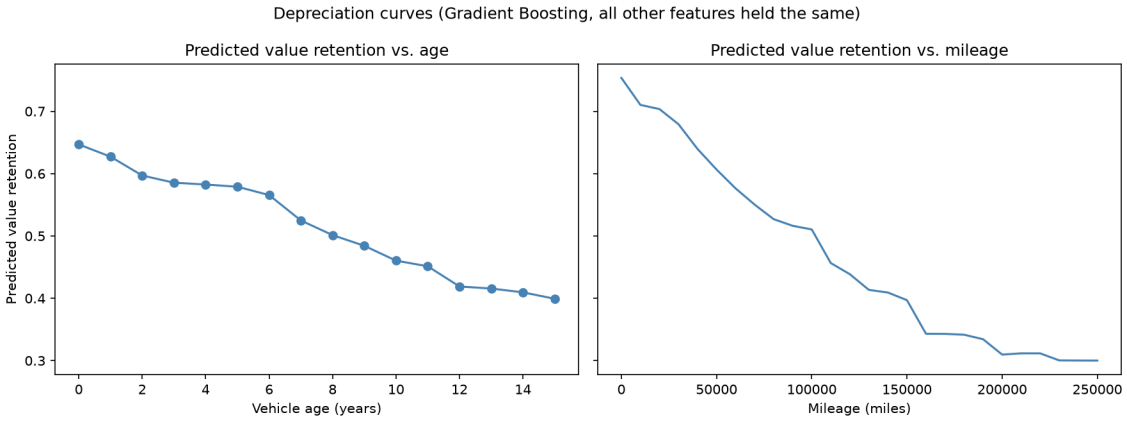
Linear regression coefficients. Fitting linear regression on unscaled features gives each effect in real units, holding all other features constant:

Change	Effect on value retention
+1 year of age	-0.023 (about 2.3% of MSRP)
+10,000 miles	-0.019 (about 1.9% of MSRP)
+$1,000 of original MSRP	-0.004
Branded title (rebuilt, salvage or lien)	-0.103
Private owner instead of dealer	-0.036
Listed on Craigslist instead of Autolist	-0.025



A branded title has the largest effect per car, about 10% of MSRP, but only about 3% of cars have one. More expensive trims lose value slightly faster. The description features have small coefficients, and some point in unexpected directions (for example, "needs work" is slightly positive and "clean title" slightly negative). Because the description features did not consistently improve the model (Section 7), these small coefficients are likely noise rather than real effects.

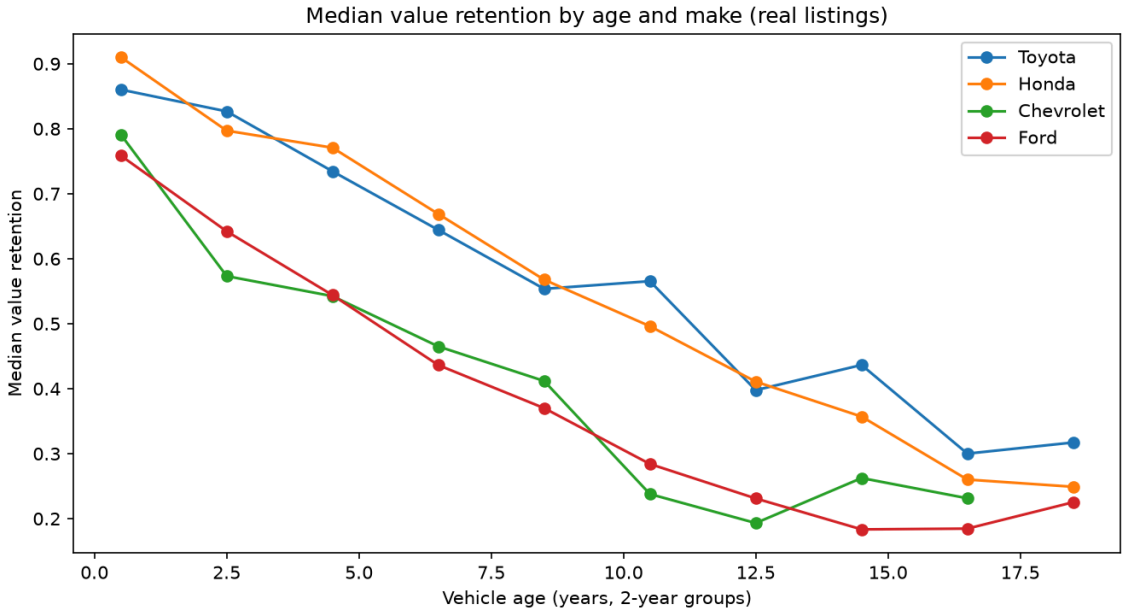
Depreciation curves. To see the shape of depreciation, I set every test car to the same age (or mileage), kept all its other features, and averaged the model's predictions.



By mileage, predicted retention falls from 0.75 at 0 miles to 0.51 at 100,000 and 0.31 at 200,000, then levels off. By age, it drops quickly in the first 2 years (0.65 to 0.60), flattens between years 3 and 6, then declines steadily to about 0.40 at 15 years.

Depreciation by make. The real data (median retention by age) shows a clear and lasting gap between the Japanese and American brands:

Age (years)	Toyota	Honda	Chevrolet	Ford
0–1	0.86	0.91	0.79	0.76
4–5	0.73	0.77	0.54	0.54
8–9	0.55	0.57	0.41	0.37
12–13	0.40	0.41	0.19	0.23



By model, median retention ranges from 0.68 (Corolla) to 0.34 (Equinox). The Corolla, Civic and RAV4 hold their value best, while the Malibu, Explorer, Escape and Equinox lose value fastest.

7. Discussion

Most important factors affecting vehicle value

1. **Mileage** is the strongest predictor. It is more precise than age, because two 8-year-old cars can have 40,000 or 160,000 miles.
2. **Age** is second. Older cars usually have more miles, so the two are correlated, and when both are in the model, mileage gets most of the credit.
3. **Model and make** come next. Toyota and Honda keep roughly 15 to 20 more percentage points of MSRP than Ford and Chevrolet at the same age.

4. **Smaller effects** include original MSRP, body type, seller type and title status. A branded title has a large effect on a single car but is rare.

Major drivers of depreciation

- **Use and time.** Mileage and age drive almost all depreciation: about 1.9% of MSRP per 10,000 miles and 2.3% per year, each holding the other constant.
- **Brand reputation.** The gap between Toyota/Honda and Ford/Chevrolet appears in the first 2 to 3 years and never closes, likely reflecting their reputation for reliability.
- **Vehicle type.** Domestic sedans and SUVs (Malibu, Equinox, Escape, Explorer) lose value fastest. Trucks and Japanese compacts lose value slowest.
- **Damage history.** A branded title costs about 10% of MSRP on its own.
- **Seller type.** Private-sale cars are listed lower than dealer cars, since there is no dealer markup or reconditioning.

Challenges in data collection and integration

- **Bot blocking.** 11 of the 14 listing sites I tested blocked automated requests with 403 errors or bot-check pages: Cars.com, CarGurus, CarMax, Edmunds, Autotrader, TrueCar, Carfax, AutoNation, eBay, CarsDirect and EchoPark. Facebook Marketplace requires a login.
- **Terms of use.** Craigslist, Autolist and J.D. Power allow these pages in robots.txt but forbid scraping in their terms of use. I kept the request volume small and slow for this class project, and the data is not shared publicly.
- **Hidden data formats.** Autolist loads its listings from a JSON API, and J.D. Power stores prices inside escaped JavaScript, so I had to inspect the raw responses instead of the visible HTML.
- **Unreliable sources.** J.D. Power sometimes returned random 403 errors, which required retry logic and a second pass. MotorTrend showed 2025 prices on its older model-year pages, so it couldn't be used.
- **Integration.** The sources used different column names, formats and category labels. Trim names didn't match between listings and MSRP data, and Craigslist model names were free text.
- **Uneven coverage.** Only Craigslist has descriptions, condition and title status, and only Autolist has MPG and interior color.

Limitations of the dataset

- **Size and scope.** 1,541 cars from 12 popular models, model years 2008 and newer (no MSRP data before 2008), and only 10 Craigslist cities.
- **Asking prices, not sale prices.** Listing prices are what sellers hope to get, not what cars actually sell for.
- **One point in time.** All listings were collected in October 2026, so the data can't show how the market changes over time.
- **No inflation adjustment.** Older cars' ratios compare today's prices to MSRPs in older dollars, which makes old cars look like they kept slightly more value than they really did.

- **MSRP matching.** 29% of cars used the model-year median MSRP because the trim was missing or didn't match. Special versions were compared to the regular model's MSRP, giving ratios above 1.
- **Source differences.** Craigslist cars are older and cheaper, so the `source` and `state` features partly measure the type of car rather than the effect of the site or location.
- **Missing values.** Condition (69%) and title status (54%) are often missing, and descriptions exist for only 47% of cars.
- **Possible bad prices.** Some very cheap Craigslist listings may be down-payment ads or scams.

Limitations of the models

- **Description features barely helped.** In the full model, shuffling any single description phrase lowered R^2 by 0.001 or less. Because Autolist cars have no descriptions, their 0s mean "no text" rather than "not mentioned". The `source` feature already tells the model which cars have text, so I also tested on Craigslist cars alone (720 cars, 5-fold CV). Adding the description features raised R^2 from 0.697 to 0.702 for Linear Regression and from 0.686 to 0.695 for Gradient Boosting. A fold-by-fold comparison (3 shuffles × 5 folds) showed the features won only 11 to 12 of 15 folds, with a mean gain smaller than its own variation. The gain is small and within the cross-validation noise. Simple keyword matching also misses negation ("not one owner") and different wording.
- **Correlated features.** Mileage and age are correlated, so their separate effects are hard to split, and linear coefficients for correlated features can be unstable.
- **Redundant columns.** The make columns are exact sums of the model columns, and mileage is on a very different scale from the 0/1 columns. Without standardizing the features, Linear Regression ran into rounding problems and its test R^2 dropped from 0.788 to 0.785, so the model keeps the scaler.
- **Small leak in cross-validation.** The state target encoding was computed once on all training rows rather than inside each fold, so the cross-validation scores are slightly optimistic. The test scores are not affected.
- **No hyperparameter tuning.** Each model used fixed, reasonable settings.

Future work

- Add more sources, older model years and inflation-adjusted MSRP.
- Decode VINs to get exact trims and improve MSRP matching.
- Use better text features (such as TF-IDF with negation handling).
- Tune hyperparameters with grid search, and compute the target encoding inside each cross-validation fold.
- Collect sold prices instead of asking prices.

8. Conclusion

This project built a complete pipeline, from scraping two listing sites to trained models, that predicts used car value retention well and explains what drives depreciation.

- **Accurate predictions.** Gradient Boosting predicts value retention with a test R^2 of about 0.82 and an average error of about 7% of MSRP (about $2,100), compared to 19% for a model that always predicts the average.

- **Mileage and age dominate.** They explain most of the depreciation: about 1.9% of MSRP per 10,000 miles and 2.3% per year, each holding the other constant.
- **Brand matters.** Toyota and Honda keep about 15 to 20 more percentage points of MSRP than Ford and Chevrolet at the same age, and the gap lasts the life of the car.
- **Description keywords barely helped.** Even on Craigslist cars alone, the gain stayed within the cross-validation noise, which suggests simple keyword matching isn't enough to get value out of seller descriptions.

The biggest improvements would come from collecting sold prices instead of asking prices, matching exact trims to MSRP, and adding more sources and model years.

9. AI Usage

Claude AI was used as a helper for when I was struggling with code debugging. However the rest of the code was mainly derived and inspired by work done in class. All the sources and data were reviewed and tested by me.